CHAPTER 9

Non-Standard Monetary Policy

9.1 QUANTITATIVE EASING

Quantitative easing (QE) is the best known but far from the only non-standard form of monetary policy. In conventional monetary policy frameworks, the operational policy target is an interest or foreign exchange rate and the balance sheet of the central bank reacts, through monetary policy operations, passively to changes in market supply and demand to bring the market clearing rate in line with the target.

Quantitative easing can be understood as a shift to operational targets for specific quantities related to the balance sheet of the central bank, such as bond holdings or base money. Employing non-standard tools is appropriate when policy action is necessary but the standard tools do not work, for instance when interest rates have reached a lower bound or no longer stimulate aggregate demand as expected [65].

Seen in this way, QE is simply the continuation of standard monetary policy by other means. The extreme nature of QE is then simply a product of the extreme circumstances in which the central bank finds itself. Standard monetary policy tools are supposed to work in a wide range of usual circumstances. A central bank resorting to non-standard tools then must be operating in a very non-standard set of economic conditions. That being said, Figure 9.1, using Japanese data, shows that central bank balance sheet expansion is not necessarily a substitute for lower rates. The chart has been formatted suggestively so that the start of the 2001–2006 quantitative easing episode appears as a continuation of the rate cuts that started in March 2001. The reduction of excess reserves in 2006 also appears like a precursor to the rate hike that occurred in July of that year. However, the response to the financial crisis that started in 2007[1] did not follow this simple narrative. Excess reserve holding increased around the time of the first rate cuts as a result of liquidity hoarding. The central bank adjusted operational targets for liquidity-supplying operations to accommodate such behaviour. In 2016, when the Bank of Japan cut the short-term interest rate target to negative (with an introduction of a tiered deposit system), the tool of excess reserve expansion had been used extensively for two years.

[1] In due fairness, the tightening in 2006 was seen at the time as premature by some observers. The decline of nominal GDP shown in the chart, which began well before the onset of the global crisis, would then not be entirely due to external shocks but may have had an endogenous component.

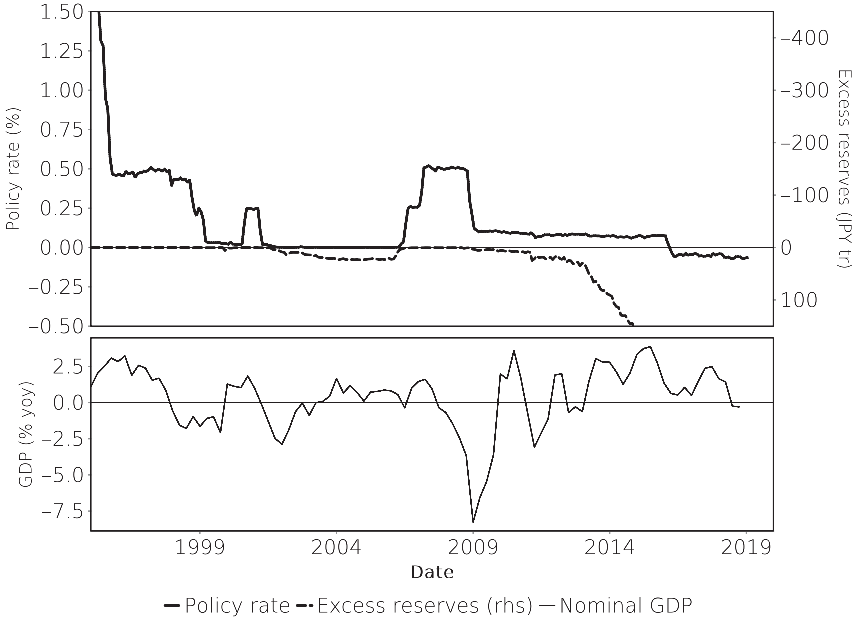

FIGURE 9.1 Monthly averages of the Japanese yen unsecured overnight (mutan) rate and excess reserve holdings at the Bank of Japan. The annual rate of change in the nominal GDP is shown for reference. Sources: Bank of Japan, Cabinet Office.

An interesting example of the distinction between quantitative easing and mere accomodation of market fluctuations was the introduction of T-bill purchases by the Fed in autumn 2019. These purchases amount to an injection of central bank liquidity into the banking system similar to the outright purchases of Treasuries that were undertaken as part of various QE earlier programmes. However, the purchases were a passive reaction to an increased demand in the US repo market, not a change in monetary policy stance. The purchases run in parallel with repo operations. Because T-bills have short maturities, the time frame over which the purchases inject liquidity into the market is longer than that of a normal repo operation but shorter than a Treasury purchase.

Just as, in Tolstoy's words, all happy families are alike, but each unhappy family is unhappy in its own way, central banks operating with a standard tool set operate in similar and well understood ways; central banks employing non-standard tools find themselves in more unique circumstances. Although experience with QE has been gathered in many currency areas, they all differ in what are probably important parameters, such as importance of the banking system, current account balance, and external debt position. Copying from other countries is therefore unlikely to succeed. Constructing a theory of QE from first principles is an alternative and more promising approach.

Quantitative easing in the monetarist sense becomes useful when the breakdown is in the link between interest rates and the money supply (II). The central bank can in such situations simply attempt to bypass the bank money creation process and control M

more directly by creating more base money. This idea was the intellectual basis for the first QE programme conducted in Japan 2001–2006. At the time, the BoJ targeted base money in the form of the excess reserves held by Japanese banks in their BoJ accounts, eventually reaching a volume of around JPY 29tr.

It should be noted that it is difficult for a central bank to assess breakdowns in the monetary transmission mechanism in real time because the quantitative nature of that link is not static. Every monetary policy action causes a reaction in the money supply (which may be zero). The best a central bank can do is to predict that reaction with some model. When the actual outcome deviates from the prediction, the reasons could be that:

- the model is incorrect or a structural change means that the model parameters no longer correspond to the new market regime;
- while the model is correct, time lags and inevitable forecast and measurement errors lead to a temporary deviation that will eventually mean-revert;
- the money market has changed in such a way as to make it unreactive to monetary policy.

Only the last situation could be characterised as a breakdown in the monetary transmission mechanism. Whether this warrants other measures by the central bank is not immediately obvious. In modern democracies the central bank is an agent of economic policy. When this particular agent can no longer work with the set of tools allocated to it, the implications are wider than a simple change of strategy at the agent[2]. At the same time, democratic processes do not lend themselves to timely adjustments of the tool set of the agent.

While quantitative easing, in the sense of increasing M by central bank fiat, is immediately meaningful in a monetarist interpretation, it would be completely meaningless to a Keynesian[3]. To understand why a central bank might want to conduct quantitative easing even when it is not monetarist, other possible channels of operation must be considered. Below is a listing of channels through which asset purchases could affect the price level.

Quantity effect (A) As outlined in Equation (6.2), more money should, *ceteris paribus*, increase the price level.

FX market (B) More supply of the domestic currency should devalue it in the FX market and thereby push up import prices. Assuming the economy contains a price-competing export sector, higher external demand for these products should also support higher inflation. See Figure 9.3 for some historical data.

[2]Some central bankers have gone as far as advocating higher inflation targets to reduce the probability of reaching the lower bound on interest rates in low growth environments. This is an alternative expression of the same dilemma. An agent is given tools to achieve an aim. When the tools do not suit the target, or vice versa, the decision to change either lies with the principal, not the agent.

[3]A major obstacle to the understanding of the monetary transmission mechanism in a Keynesian/neo-classical setting is that the related economic models tend to ignore the role of the financial sector in the economy.

Signalling (C) By using its potentially unlimited balance sheet for asset purchases, the central bank signals strong commitment to its price or inflation target.

Relative pricing (D) By intervening in asset markets, the central bank affects the relative pricing of different capital sources in the economy. This can stimulate some sectors relative to others, or subsidise some forms of lending relative to others.

Portfolio effect (E) Because the central bank removes assets from the portfolios of private investors, these investors must reallocate cash to other assets. Even though the central bank itself does not target these alternative assets (as in the relative pricing effect above), they benefit from the spillover of the extra liquidity.

Bank capital formation (F) By increasing asset prices, the central bank can create permanent new equity capital in banks because it allows them to sell assets above amortised cost.

Risk-weighted assets (G) If the central bank buys assets with non-zero risk weights from banks, it frees up existing bank equity.

Lower Volatility (H) Central bank buying can dampen the price volatility in the affected asset markets. This can free up bank equity (via lower VaR measures), flatten the yield curve (due to lower term risk premia), and stimulate investment (due to more certain present values of future cashflows). Because all these effects are indirect and already listed above, these are not separate operational channels of QE. It should also be noted that prolonged low volatility can be detrimental to financial stability. While some investors will increase leverage to achieve the same risk-adjusted returns, banks have to disinvest from their market making operations. Should volatility then return due to some external shock, large deleveraging would occur in an environment of depleted market-maker balance sheets.

Pure monetarists would agree only with the first two channels (A) and (B), and also disagree with the second sentence in (B). A more Keynesian interpretation of QE would deny (A). Point (C) was made by the Bank of Japan in its explanation of the current QE programme. The ‘relative pricing’ (D) aspect is most prominently reflected in the Fed’s ‘Operation Twist’, which was targeted directly at lowering long-term interest rates, and the Fed’s purchases of mortgages, as well as the Bank of Japan’s yield curve control (YCC). Any such thinking would meanwhile be anathema to an ordoliberalist economist[4].

The two channels (F) and (G) are particularly important for markets where the breakdown in the control relationship (II) is caused by a lack of bank capital and where banks provide the bulk of funding to the real economy. That said, central bank policy that is explicitly geared to supporting bank capital levels would run counter to the general principle that the role of a central bank is the provision of liquidity, not of capital[5].

[4]This school of economics is not structurally tied to either monetarism or the neo-classical school but its representatives tend to be monetarists.

[5]The standard prescriptions of central bank crisis tools remain Walter Bagehot’s exhortations to the Bank of England [5]: ‘that in time of panic it must advance freely and vigorously to the public out of the reserve [. . .] and the advances should, if possible, stay the panic. And for this purpose

While some politicians and economists view the FX channel as important, the evidence for the stimulative effects of a weaker currency on exports depends on the nature of industry and non-price competitiveness. It is quite possible that FX depreciation simply creates windfall benefits for highly competitive companies while companies without a viable growth model are unable to enjoy improved cost competitiveness.

The variety of possible channels of QE into the real economy means that it is difficult to specify a required size for such a programme before specifying what mechanisms and assets it should target. A purely monetary interpretation of QE would imply sizes that are meaningfully large relative to the existing money stock, and by implication, GDP. A more Keynesian programme targeted at specific asset classes could operate on sizes that are large relative to the pool of eligible assets yet small compared to the total money stock (and indeed would not be seen as quantitative easing by a monetarist observer for that reason). It is therefore useful, at least for a non-monetarist observer, to distinguish between quantitative easing (aimed at addressing a breakdown in the relationship (II) above) and asset purchases (which may have more specific targets). Note that the Bank of Japan conducted quantitative easing with an emphasis on repos, not asset purchases, between 2001 and 2006. In practice, the ability of central banks to adjust purchase programmes during their implementation means that the inevitable forecasting errors regarding their impact does not diminish their justification. The forecasts need to be accurate enough to weigh the advantages and disadvantages of conducting such operations, versus the available alternatives, with a reasonable degree of certainty.

9.1.1 The Monetary Effect of Large-Scale Asset Purchases

When central banks purchase macroscopically large amounts of assets, the impact is not limited to the banking system, i.e., the direct counterparties of the central bank, as would be the case with standard operations. Banks looking to sell assets to the central bank have to source them from other asset owners, such as funds, insurance companies or households. This creates a conceptual problem because these actors are unable to hold directly the asset created by the central bank in exchange, namely new central bank liquidity. Instead, the asset sellers merely substitute these assets with claims against banks. This means that the banking system in aggregate acts not only as an intermediary in the asset transfer, but is forced to lengthen its aggregate balance sheet to accommodate these new claims.

Figure 9.2 demonstrates this graphically. In economic terms, the increase in outside money (the quantitative easing effect) is matched by an increase in inside money.

This mechanism means that asset purchases potentially have an additional side effect. If the security purchased by the central bank is of very high credit quality, a simple bank deposit (which exposes the depositor to the credit risk of the bank) may not be an acceptable substitute. The asset manager may instead ask for the deposit to be collateralised, i.e., a repo transaction. While in principle this repo only amounts to swapping

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there are two rules: First. That these loans should only be made at a very high rate of interest. [. . .] Secondly. That at this rate these advances should be made on all good banking securities, and as largely as the public ask for them. [. . .] No advances indeed need be made by which the Bank will ultimately lose.'

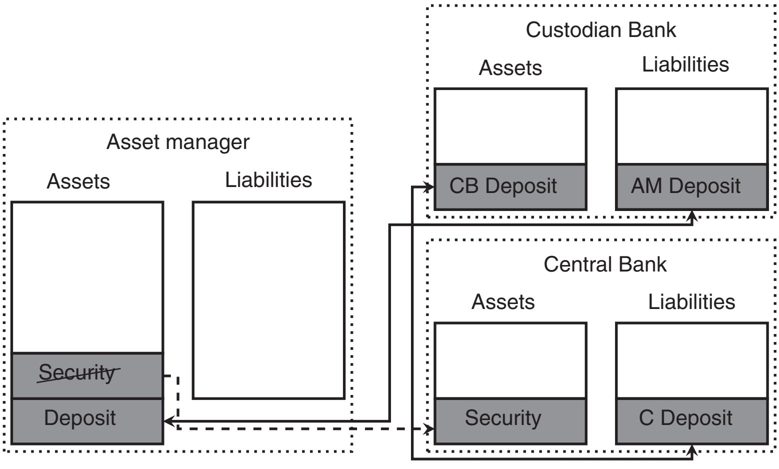

FIGURE 9.2 Balance sheet effect of large-scale asset purchases. Corresponding assets and liabilities are paired by arrows; the dashed arrow denotes the transfer of the security purchased by the central bank.

central bank deposits against securities elsewhere in the system, the repo market may not at all times be efficient enough to accommodate these transactions without friction costs. These friction costs may explain in part why general collateral repo rates can decline as a result of large-scale asset purchases.

9.1.2 Market liquidity and central bank asset purchases

The modern asset management industry is driven by market indices which are used either as performance benchmarks, or strict asset allocation guidelines, depending on the amount of tracking error targeted by a given asset manager. Exchange-traded funds (ETFs) in particular are closely tied in their asset allocation to a given bond index. The index weight of a given bond is given simply by the market value of the asset, divided by the sum of all market values. An index-neutral asset allocation would therefore be one where each asset is held in direct proportion to its outstanding volume.

In practice, such an index-neutral asset allocation is impossible to realise because there is a structural demand preference for certain assets by investors who do not follow a broad market index. Foreign reserve managers at central banks, for instance, tend to prefer the most liquid assets regardless of expected returns. This is because the main concern for such portfolios is not their return, but the ease with which they can be turned into cash in times of crisis. Other investors, such as insurance companies, follow investment strategies that are based on particular return targets linked to liabilities. At times of lower market yields, such portfolios purchase riskier assets.

Central bank asset purchases can exacerbate this problem even when they follow market weights. As an example, assume a segmented market consisting of two types of asset, each of which comprises 50% of the total market volume. One asset is of very high quality and 40% of the asset is held by investors who are constrained to hold such high quality assets. The other asset is perceived as more risky and consequently no forced holders are present in this asset. As a result 60% of the first, and 100% of the second asset are held by value-oriented investors that can buy or sell depending on market prices. If the central bank now steps in and buys 30% of the total market volume in proportion to the outstanding amount, it will reduce the free float of the first asset by half but the free float of the second by less than a third. The price impact of the purchases is therefore likely to be larger for the first asset.

Central banks can, and do, mitigate the distorting effects of such local imbalances in various ways [25]. For instance, large purchase programmes have securities lending facilities so that the availability of special repo collateral is not negatively affected by central bank holdings. Purchases are also conducted in ways that avoid excessive security-specific price effects. Depending on the central bank, the way in which this is done varies. For central banks using auctions, some form of relative value analysis takes place when offers are evaluated. Others may incorporate such analysis in bilateral operations. The way in which implementation frameworks across central banks differ does not only apply to the mode of purchase, but also to information disclosure about purchases. The experience from large-scale asset purchases around the world is that there are multiple optimal combinations of purchase mode, flexibility of purchase, and information disclosure. Which of these optima is chosen depends on local market usances. In every case, however, central banks have made adjustments to their programme implementation in order to protect market functioning. Doing so requires a large amount of market monitoring and analysis.

9.1.3 Helicopter money

Quantitative easing as described here remains a monetary policy tool in the sense that it provides an aggressive, but temporary, injection of liquidity into the economy. The next escalation step could therefore be to make the liquidity injection permanent. At that stage, however, monetary policy crosses into the fiscal policy realm because the idea of controlling the money supply in order to preserve the store of value property of money would be subordinated to other aims, such as dealing with a debt overhang. By permanently expanding the money supply, the central bank would engineer a jump in the price level, according to Equation (6.2).

The most graphic description of this approach is the idea of dropping banknotes from a helicopter, put forward by Milton Friedman in 1969. As Ben Bernanke pointed out in 2002, the same concept could be implemented by fiscal measures that are funded by monetary expansion[6].

What appears to be misunderstood is that this approach is actually very straightforward and to some degree already implemented. Some authors (e.g. Nyborg [64])

[6]The unsung hero of economic education, Douglas Adams, popularised this concept in 'The Restaurant at the End of the Universe' [1] as follows:

> 'How can you have money,' demanded Ford, 'if none of you actually produces anything? It doesn't grow on trees you know.' [. . .] 'Since we decided a few weeks ago to adopt the leaf as legal tender, we have, of course, all become immensely rich.' [. . .]

make the assumption that helicopter money requires the explicit cancellation of government debt by the central bank, or the conversion of debt held by the central bank into infinite maturity zero coupon bonds. The actual mechanism can be much simpler. Most developed nation central banks are required by law to remit any profits over a certain threshold to their respective governments. Coupon payments by the government to the central bank are therefore ecomically neutral because they will be remitted after the end of the fiscal year. The level of coupons held by the central bank is therefore more or less irrelevant, aside from pull-to-par effects. In addition, the maturity of a government bond is not equivalent to a fiscal contraction simply because the redemption amount will normally be covered by the issuance of a new government bond. Instead of accepting perpetual bonds, a central bank can simply commit, implicitly or explicitly, to reinvesting any maturing principal in new government bonds. In other words, the transition between quantitative easing and helicopter money need not be tied to the shape in which the initial liquidity injection takes place, or the types of bond acquired by the central bank. Instead, it is dependent on the evolution of central bank involvement in the sovereign debt market.

The Bank of Japan is a useful example. It has purchased Japanese government bonds outright since at least 1967 [81]. To the extent that the volume of government bonds held by the BoJ never fell to zero, these purchases are economically indistiguishable from helicopter money. A debt that has been rolled over for in excess of 50 years is simply not very different from a perpetual asset. That being said, the Bank of Japan until 2013 applied a 'banknote rule' that limited the amount of JGB it could hold to the total volume of banknotes in circulation. In doing so, it limited the amount of 'helicopter money' roughly to the amount of seigniorage remitted to the country. According to the US Federal Reserve, outright holdings of US Treasury assets have over the last 15 years also never fallen below USD 500bn. These 500bn, equivalent to about 2.75% of current nominal GDP, are also hard to separate from helicopter money. To put this into some context, foreigners hold physical US dollars outside the US without the intent to present them for payment in the US itself. The seigniorage associated with this money accrues to the United States. A report by the St. Louis Fed [4] gives an estimate of USD 500bn for such notes, which is the same amount as that contributed by the Treasury holdings of the Federal Reserve. While both factors create a fiscal boost, the monetary of these two factors is subtly different because helicopter money adds to domestic liquidity while notes held abroad diminish it[7].

> 'But we have also,' continued the management consultant, 'run into a small inflation problem on account of the high level of leaf availability, which means that, I gather, the current going rate has something like three decidious forests buying one ship's peanut.' Murmurs of alarm came from the crowd. The management consultant waved them down. 'So in order to obviate this problem,' he continued, 'and effectively revalue the leaf, we are about to embark on a massive defoliation campaign, and ... er, burn down all the forests. I think you'll all agree that's a sensible move under the circumstances.'

[7]Indeed the purpose of estimating the volume of physical currency held abroad is to be able to counteract this contraction of domestic liquidity.

In any case, the liquidity effect of the long-term holdings of central banks has so far been minuscule, and certainly too small to have an impact on inflation. In this sense, the practical test of helicopter money has still not occurred.

9.1.4 Choice of methods and assets

The choice of how to conduct quantitative easing and what assets to involve in the implementation depends on the central bank's assessment of how QE works. A purely monetarist central bank looking at effects (A) and (B), or even (C) above can choose freely between repo or outright transactions, and can use any type of financial asset, or indeed non-financial asset, in these operations. The hurdle for the purchase of non-financial assets in the monetarist view is that it influences real economic activity (dY) too much whereas a Keynesian would see the role of the central bank as easing financing conditions for the government to conduct such purchases.

When the central bank takes a more Keynesian view of quantitative easing, the choice of operation becomes very skewed towards outright purchases. While repo operations provide funding and signal central bank intent, the effects linked to portfolio shifts and bank equity rely on outright purchases. Last but not least, if the central bank is targeting bank equity levels (implicitly or explicitly), it must remove assets from bank balance sheets in such a way that they can be de-recognised in accounting terms.

Assuming outright asset purchases are chosen as the implementation method, the choice of assets becomes more important for a number of reasons. First, a central bank choosing to act via relative pricing (D) needs to purchase the relevant assets directly and assume a considerable risk of default; if it is confident that portfolio shifts (E) will be sufficient, it can target other asset classes where, for instance, asset purchases are less distorting or technically easier. Also, the credit risk of outright purchases is much higher than on repo operations with daily margining. The central bank must therefore weigh the cost of credit risk against the benefit of targeting specific asset markets.

With this in mind, one can discuss possible asset classes for outright purchases in terms of the potential effects listed above. Below is a list of assets and a summary of their potential effects in a simple matrix.

Government bonds Due to their superior liquidity, government bonds are the natural choice for executing large-scale asset purchase programmes. Lower government bond rates can also affect funding costs for other actors in the economy such as banks and high-grade corporates. That said, even deep government bond markets can experience low liquidity as a result of central bank asset purchases if these are to be large relative to GDP. Because repo operations would achieve the same monetary expansion goals, such purchases may call into question central bank independence beause future central bank policy may become constrained by the default risk of a large government bond portfolio (this situation is known as fiscal dominance).

Supras, Subsovereigns, Agencies Depending on the market, there may be near-substitutes for government bonds available. Aside from the US agency MBS market, in no market are such securities comparable size and liquidity with government bonds. Purchases of SSA securities are therefore most useful via portfolio allocation shifts.

Covered Bonds Covered bond purchases, in those markets where these bonds are high in volume are a natural extension of repo along the curve because covered bonds also are standardised collateralised bank obligations. However, liquidity in covered bonds is substantially lower than for government bonds in all markets aside from Denmark.

Senior unsecured bank debt Bank debt is an abundant asset class but would pose a risk management dilemma for central banks that use repo as a liquidity provision tool. Holders of bank debt carry credit risk of the issuing bank. Doing so on an unsecured basis for a longer term whilst asking for repo collateral on short term operations is logically inconsistent. Furthermore, any such purchase would require taking a view on spreads which would invariably call into question the impartiality of the central bank vis-à-vis its counterparties, especially where the central bank has superior insight into the situation of each bank issuer.

Corporate debt Corporate bonds are a large asset class but are illiquid in most markets. More concerning for an asset purchase programme is that debt financing is more easily available for companies with a large stock of fixed assets and stable cashflows. High-growth start-up companies operate on very low leverage ratios because they are less able to source debt financing. A breakdown of major bond indices shows a large share of utilities (telecommunications, energy, gas), mining, transportation and car manufacturing companies among the issuers of investment grade bonds. Lowering funding costs for such companies is unlikely to be growth positive which would be a pre-condition for a Keynesian QE approach. Corporate bond purchases also implicitly discriminate by size against smaller and unrated companies. That being said, funding larger corporations through the central bank can free up bank balance sheets for the loan funding of smaller ones.

Bank loans Banks lend under slightly less onerous conditions (albeit potentially at higher rates) than the public markets because they operate under less information asymmetry than bond investors and may have access to collateral. Purchasing bank loans would therefore be more stimulative for high-growth companies than the corporate bond market. Additionally, it would have a beneficial effect on bank equity via RWA reduction and potentially capital gains if banks have loans to sell at book value or above (banks are unlikely to sell loans below book value). However, a purchasing central bank faces a selling bank counterparty with superior information about the loan asset and probably has insufficient analysis capabilities to value large numbers of loans across different jurisdictions.

The only realistic way of purchasing bank loans is therefore likely to be via securitisation where loan pooling and tranching provides a sufficient risk buffer for the purchasing central bank.

Foreign assets A way of focusing QE in the FX channel would be the direct purchase foreign assets without sterilising the resulting liquidity. The structural problem with unsterilised foreign asset purchases is that the targeted amount of FX intervention and increase in the money supply are rigidly coupled. In addition, the central bank could be criticised for taking credit risk in foreign markets, and for spending resources in a way that might benefit foreign economies via the QE channels (D) through (G) above while these effects would be absent for the domestic economy.

In practice, a more serious concern would be that pursuing the FX channel in such an overt fashion would be seen by other countries as unfair.

Equities The Bank of Japan has been purchasing equities through exchange-traded funds (ETFs). Japan is somewhat unique in that it has a large number of listed small companies (the Topix index has nearly 2,200 constituents and the index weights have a Herfindahl index of 0.0064, meaning it is well diversified). For Europe, the points made on corporate bond purchases would also apply to listed equity.

9.2 PRACTICAL EXPERIENCE

9.2.1 QE, money multipliers and FX

The experience of early quantitative easing programmes has been that it is hard to prove efficacy along the lines in which they were designed to act. While the expansion of central bank balance sheets is directly observable, the evidence on broader money aggregates is less convincing. As Figure 9.3 shows, money multipliers dropped during the Japanese and US quantitative easing programmes (essentially suggesting that Equation (6.4) may have held) which raises doubts over the ability to bypass the standard private money creation mechanism[8]. The evidence in the euro area is more mixed in that the money multiplier declined before large-scale balance sheet expansions were undertaken[9].

Of course, the problem with such analyses is that they are by definition relying on counterfactual thinking. It is quite possible that broad money aggregates would have declined if central bank balance sheet expansion had not taken place. Monetary policy loosening is a response to a decline in economic activity and such a decline is likely to coincide with a drop in private money creation. As always in macro-economic problems, the option of conducting experiments with a different policy path but otherwise identical conditions is not available. Furthermore, the idea that a QE programme can have a signalling purpose very much negates the idea of an ex-post quantitative analysis of efficacy. Signalling effects are by definition circular: they work because they work.

As mentioned above, money velocity cannot be measured directly because many monetary transactions are unobservable (or at least unobserved by statisticians). As a proxy, one can use the payments volume in central bank reserves transfer systems. Since physical cash payments are only a small part of total payments in most economies, the interbank transfers of central bank reserves may capture a broader subset of total payments because they are often reflective of customer payment orders.

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[8]M2 is chosen here because it can be found for all jurisdictions, not because it is the most appropriate money aggregate. The Federal Reserve stopped the publication of M3 as of 23 March 2006.

[9]That being said, the transition to fixed-rate full allotment of ECB main refinancing operations, and the narrowing of the corridor between marginal lending facility and deposit facility can be viewed as a passive balance sheet expansion.

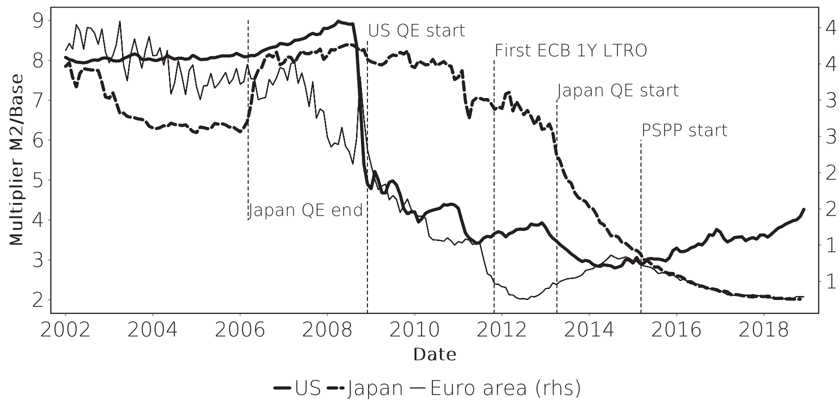

FIGURE 9.3 Money multipliers (M2/Base money) in the US, Japan and the Euro area. Sources: FRED, Bank of Japan, ECB.

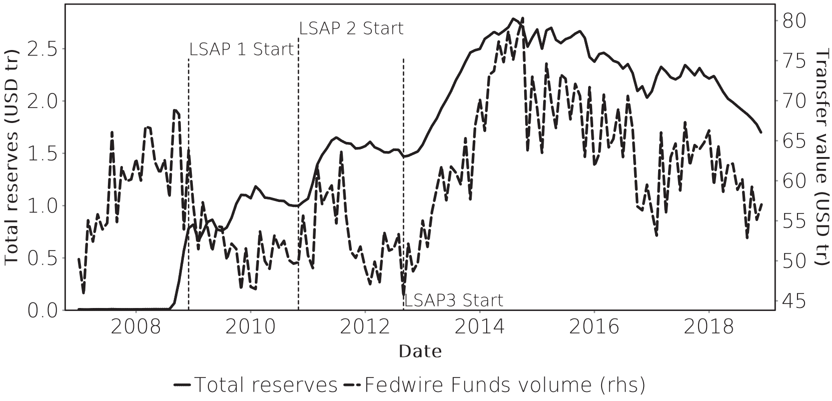

FIGURE 9.4 US total reserves and monthy Fedwire Funds transfer values. The relationship appears to be strenghening after 2013. Sources: FRB Services, Federal Reserve.

Such reserve transfers systems are operated by central banks (Fedwire Funds by the US Federal Reserve System, Target2 by the ECB) or private bank associations (the Zengin System[10] by the Japanese Bankers Association's Zengin-Net, Faster Payments by Faster Payments Scheme Limited in the UK).

A study by the ECB on the liquidity velocity of central bank money in the Target 2 system (Box 3 in [7]) found a significant decline in money velocity as base money increased although the relationship does not appear to be a straightforward case of Equation (6.4). The ECB does not disclose the underlying payments volumes but such disclosure is available for Fedwire Funds.

[10]Zengin is the Japanese short name for the Japanese Bankers Association. The Zengin System is formally called the Zengin Data Telecommunication System.

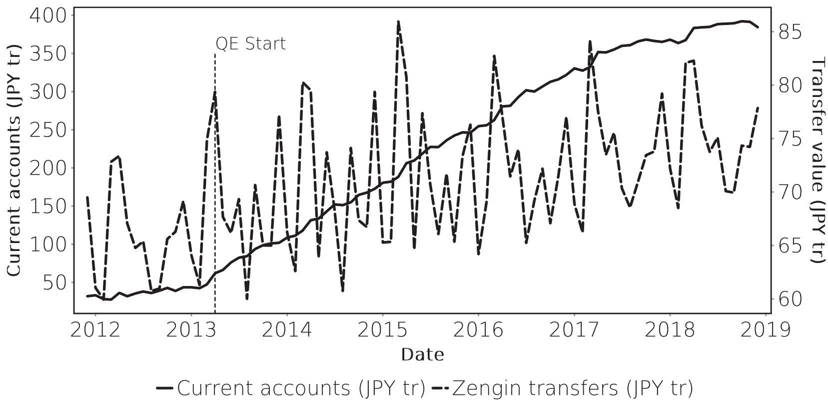

FIGURE 9.5 Monthly average current account holdings of Japanese banks at the BoJ and total small (under JPY 100m) transfer values in Zengin. Sources: BoJ, Zengin-Net.

Figure 9.4[11] suggests that since 2013, there has been a closer relationship between total payments volume and money supply which would contradict Equation (6.4) and support the idea that quantitative easing can have an effect on economic activity. A single dollar in bank reserves turned around in Fedwire Funds 28.9 times on average between 2012 and 2018 with a standard deviation of 9.4%. Between January 2007 (the start of Fedwire Funds data availability) and the onset of monetary easing in 2008, a single reserve dollar moved 6,500 times, with a comparable 9.7% standard deviation. A more stable velocity of reserves (i.e., a small standard deviation) would suggest a higher sensitivity of prices to money supply through Equation (6.1). That being said, Fedwire Funds transfers may also be the result of trading in financial assets which are outside the scope of this equation. One could posit that quantitative easing may be stimulating the trading in financial assets and that this explains the relationship between the size of the Fed balance sheet and payments volumes.

At the same time, one could interpret higher transaction volumes in conjunction with larger central bank balance sheets globally as a sign of lower trust in private banks. To the extent that bank customers wish to reduce their exposure to banks, there will be a shift from transacting in bank money (deposits) towards transacting in securities. This then creates more demand for transactional central bank money.

The Japanese Zengin System data, shown in Figure 9.5, is more in line with Equation (6.4). Because Zengin breaks out the volume of small value transfers[12],

[11]The figure uses the official acronym LSAP for large-scale asset purchases. Note that the Fed also conducted lending operations that injected additional liquidity before the start of these outright purchase operations.

[12]Small value transfers are collected and settled in central bank money at designated times through the Bank of Japan's RTGS system. Zengin transfers over JPY 100m are settled in real time through RTGS.

one can make the assumption that such transfers are related to purchases and wage payments which are within the scope of Equation (6.1) while financial instruments transactions are less relevant. To further match the velocity measure to the underlying stock, Figure 9.5 uses only current account holdings at the Bank of Japan because these can be transferred through Zengin. Banknotes and coins in circulation, which are also part of the monetary base, are excluded. Transaction volumes since 2012 have been extremely stable with an average annual growth rate of around 1.7% and apparently unconnected to the evolution of the current account volumes.

The UK data is unfortunately not stable enough to conduct this type of analysis because interbank payments underwent a significant transformation in the last decade. The large-volume CHAPS system was replaced by the faster, and considerably cheaper, Faster Payments system and this transition affected end-user take-up of such transfer systems.

The evidence on on the FX impact of quantitative easing is mixed. Figure 9.6 shows the ratio of the money multipliers in Japan and the US, each defined as M2 divided by base money. This ratio decreases when the money multiplier in Japan falls, for instance as a result of quantitative easing, *relative* to that of the US. Because more easing in Japan could be expected to cheapen the Yen (leading to an increase in the JPY/USD exchange rate), the exchange rate is plotted on a separate, inverted axis.

Some common trends between the two series appear to exist although the correlation cannot be called strong. The relative easing of monetary policy by the Bank of Japan following the bursting of the Japanese bubble occasionally coincides with Yen cheapening but the overall level in 2006 (at the peak of the first QE) is not much weaker than in 2001. The quantitative easing in the US is clearly visible as an increase in the ratio of money multipliers, but the timing and evolution of the post-Lehman appreciation of the Yen does not seem to be fully synchronised. The more recent history again shows a better correlation. Note that FX interventions were a more frequent occurrence during the earlier part of the history shown here.

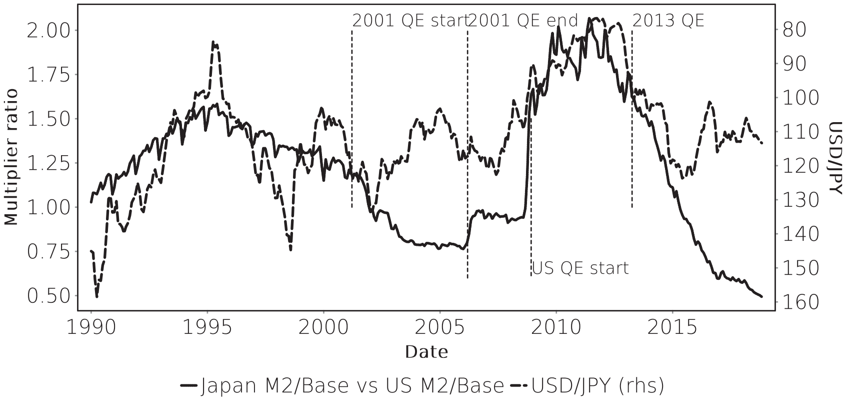

FIGURE 9.6 Long-run series of the ratio of the Japanese and US money multipliers and the JPY/USD FX rate. Sources: Federal Reserve, Bank of Japan.

The sporadic nature of the correlation between QE and FX rates is perhaps a good indication for the limits of central bank policy. Japan has been running a persistent current account surplus (trade and income) over the period shown here. Such a surplus should be positive for the currency to the extent that some of these flows are repatriated rather than reinvested abroad. A central bank can temporarily offset these long-term trends with activist monetary policy but apparently not competely decouple the FX rates from trade fundamentals. The weakening of the Yen to some degree also stems from the deterioration of Japan's trade balance following the Fukushima disaster and not only from Abenomics and BoJ policy. The observation that the Yen depreciated 23% against the USD before QE was started by the BoJ, and since then moved only 7%, is evidence of the relative importance of other factors for the FX rate.

That said, even a temporary suspension of normal price formation mechanisms could in theory be supportive for the real economy in the longer run. A central bank could look to allow capital formation in the manufacturing sector via a policy that depreciates the currency, creates extra external demand, and thus allows more profits to be invested in machinery and bank loan repayments. At the end of this process, the manufacturing sector would be less dependent on labour input and the banking system less leveraged. A re-strenghtening of the currency would then not cause any substantial problems for growth. However, the materialisation of such hysteresis effects is by no means certain.

9.2.2 Bank of Japan 2013 QE experience

The experience of the Bank of Japan's quantitative easing programme in 2013 highlights the importance of technical factors. The Bank of Japan announced on 4 April 2013 a quantitative easing programme across multiple asset classes (JGB, CP, corporate bonds, ETF, REIT) which amounted to the purchase of around 70% of the gross issuance of JGB and thereby more than the net supply of government debt.

Interestingly, interest rates increased after QE was announced which would run counter to the expectation that announcing a programme to buy a particular type of asset would make it more expensive. Over a year after the start of QE, Japanese 10Y yields (0.6%) were still higher than they were before QE was announced. Two factors can be identified to explain this phenomenon besides higher realised inflation.

The first was a communication problem. Bank of Japan Governor Kuroda was ambiguous about the exact channels in which he expected QE to affect the economy and it took some time to clarify that signalling effects were the main channel in which the Bank of Japan expected its policy to work. Because the Governor expressed indifference over the direction of yields (saying that a rise in JGB yields as an expression of higher inflation expectations was understandable), bond market participants had difficulties in interpreting the price signals the BoJ wanted to send. As a matter of fact, the BoJ did not intend to send any such signals.

The second problem was implementation. In its original communication on 4 April 2013, the operational schedule was for 11 operations to be held on 6 days of the month (meaning that the BoJ would conduct two operations on the same day). The size of each operation was to be around JPY 1tr (EUR 7bn) and JPY 300bn (EUR 2bn) in the long end. These sizes were apparently chosen in reference to JGB auctions which are JPY 2.3tr in size for 10Y JGB and JPY 700bn for 30Y JGB. While the immediate reaction of the market was positive, dealers judged these sizes as too large to be able to offer sufficient amounts of securities at each operation. This disappointment is reflected in the sell-off in JGB that followed immediately after the initial rally. The BoJ reacted to this criticism by changing the operations schedule two weeks later. On 18 April, a new schedule was announced that increased the number of operations to 17 on 8 days of the month, with a corresponding reduction in the size of each operation to JPY 700bn. This was further refined on 30 May 2013 to 19 operations to be conducted on 10 days.

Meanwhile, the spike in yields caused problems for Japanese commercial banks who are large holders of JGB for carry purposes. Because the average maturity of the carry portfolio is slightly longer than the maturity of deposits, volatility along the curve has an impact on capital levels. The higher volatility of the 2Y–5Y spread in particular caused problems for these banks.

This encouraged large banks to sell their holdings in intermediate JGB in order to reduce risk. From a monetary policy point of view, while the purchase of more than the net supply of government bonds would of course at some point require the reduction of JGB holdings in the private sector, the rapid nature of this selling, and the attendant increase in volatility, are unlikely to have been part of the planned outcomes.

By now, the Japanese market has adapted to the situation and volatility has fallen to very low levels. The realised volatility of the JGB market is now roughly 0.6% annualised which implies an average daily move of around 4.5 cents across all JGB. Trading volumes have fallen as well and about 7.5% of all trading in JGB is now directly related to either auctions or buy-backs. No single trade in the 10Y benchmark JGB was recorded on 14 April 2014[13].

9.2.3 Lessons from the initial BoJ quantitative easing

These developments raise the question what would happen in the Japanese market if volatility were to reappear suddenly. The experience of the ‘VaR shock’ of the summer of 2003 is instructive in this respect. At the time, yields and volatility had been falling for some time under the influence of gradual monetary easing by the Bank of Japan. After several years of rallying bonds, traders who would take active short positions had gradually been weeded out of the market. In this situation, US bond yields backed up substantially on the back of an apparent change in Fed policy. The rapid rise in JGB yields that started in June 2003 could not be stopped by filling short positions; instead the existing long positions at dealers came under pressure and were in addition subject to higher VaR charges. Stop-loss selling became inevitable and created a positive feedback loop that accelerated the move upwards in yields; eventually driving a second wave of selling in August 2003. Note that these events occurred during the first quantitative easing episode of the BoJ, not as a result of liquidity withdrawal.

[13]While the 10Y JGB is the bellwether of the Japanese bond market, it actually has no natural investor base, being too long for banks and too short for life insurers.

The actual withdrawal of liquidity by the Bank of Japan in 2006 did not immediately disrupt general interest rate levels. The BoJ withdrew liquidity faster than it had injected it. JGB yields increased gradually from around 1.5% to 2%. It should be noted that at the time the Japanese market was, as happens sporadically, used as an easy way to express a global reflationary views via short rates positions. The impact of liquidity withdrawal was more subtle and better observable in the sub-sovereign space.

There were two shocks in the Japanese municipal bond market that provide a clue as to what the end of QE meant for fixed income markets. The first is the Yubari shock of 20 June 2006 when the small town of Yubari in Hokkaido[14] applied for fiscal reconstruction status. The timing of this news coincided with the winding down of excess reserves at the BoJ and various types of sub-sovereign debt experienced spread widening. A second and unrelated episode of municipal stress around year-end 2007, after the liquidity levels stabilised, was much more limited and did not extend to other issuer classes.

The conclusion is that the end of QE removed the certainty of abundant funding that would normally encourage value investors to pick up assets that had declined too much in price. The widening of spreads in summer of 2006 was therefore not the result of liquidity withdrawal itself, but the result of a decline in risk appetite driven by that liquidity reduction. The market was more settled in 2007/2008 so a similar credit event had a very different and more localised impact.

Viewed in a wider context, this evidence from the unwind of the first QE programme in Japan appears to contradict a narrow monetarist interpretation of the impact of QE. Although the operational implementation of the programme used short term operations, the impact of the unwind suggests that private sector investors bridged the liquidity into asset purchases. Exit strategies from large-scale asset purchases communicated by central banks now generally take longer in order to avoid side effects of sudden liquidity withdrawals.

9.3 NEGATIVE INTEREST RATES

Until the financial crisis of 2008, negative interest rates were generally seen as aberrations that only occurred in special market segments, like the repo market for specials. Negative offer rates for deposits were seen in markets where policy rates were near zero and some banks had no desire to increase the size of their balance sheets. None of these observations contradicted the economic intuition that physical cash offers a risk-free investment at an interest rate of zero and so acts as a floor for risk-free interest rates.

As a result of the financial crisis, negative interest rates have become more prevalent for two reasons. First, policy rates declined globally and moved close to zero. In some instances, particularly the Japanese market, cross-currency funding strategies then made it economical to invest in assets at negative interest rates to generate funds in other currencies via the FX swap market [80]. The second reason is that central banks found that cash holdings even at negative policy interest rates do not increase

[14]Population at the time was around 12,000 but has since fallen to below 10,000.

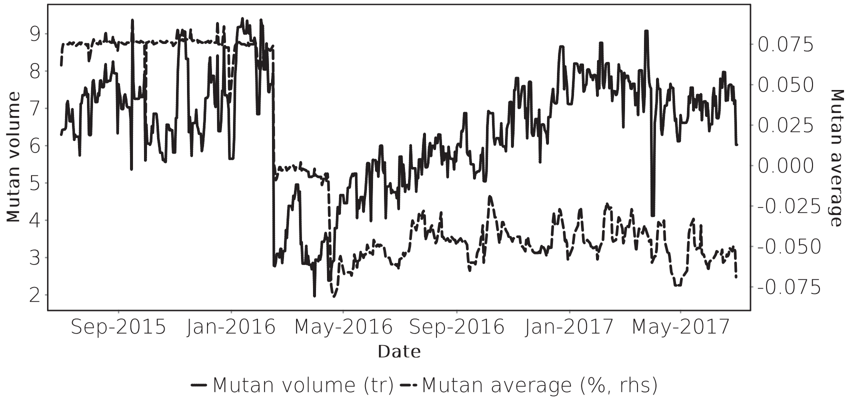

FIGURE 9.7 Japanese daily mutan volumes and average rates July 2015–July 2017. Source: Data from The Tokyo Tanshi Corporation Ltd, Japanese daily mutan volumes and average rates, © THE TOKYO TANSHI CO, LTD.

significantly. As it turns out, cash has storage costs and is more difficult to transact than central bank current account holdings. Central banks can therefore lower policy rates even below zero.

The relevance of negative interest rates is profound. In a multi-currency setting, a floor on interest rates can limit the ability of a central bank to maintain a constant level of accomodation relative to another central bank when that other central bank cuts interest rates. From a signalling point of view, an interest rate floor can lead to a market view that a central bank has limited room for accommodation even in extreme low inflation situations. Yield curve models based on short rate drifts must assign non-zero probability mass to negative interest rates and as a result should predict flatter curves[15] [65].

The transition to negative rates had different effects in different currency areas. While is has been smooth in most countries, the experience in Japan, where negative rates were introduced relatively late and without much prior warning on 29 January 2016, interbank money market volumes declined by 75% from one day to the next and took around one year to recover (Figure 9.7).

9.4 THE SPECIFIC SITUATION OF THE ECB

The decline in inflation following the financial crisis has been uneven across member states that make up the euro area. In the setting of Equation (6.2) on page 43, structural reforms (including a more efficient allocation of labour) imply positive dY and hence a negative contribution to dP. To the extent that such effects explain diverging inflation

[15]The significance of the latter effect is debatable because the impact of negative rates on expectations can be offset by other drift terms.

trajectories, they do not create a reason for monetary policy to act. Indeed, positive dY may in a Keynesian setting suggest reasons to tighten before the output gap shrinks too drastically. However, declines in inflation in some euro area countries were also due to weak domestic final demand, partly caused by fiscal consolidation.

Asymmetric inflation declines can be counteracted with targeted quantitative easing programmes and other unconventional policy measures while they are less susceptible to conventional monetary policy tools. Given the institutional constraints of the EU, symmetric fiscal policies are limited in scope, shifting the pressure on the ECB to act in such a way.

One of the features of the EU Treaties, and by implication the design of the monetary union, are the strong restrictions that are placed on the EU institutions. The sovereign nation states that make up the EU have jealously protected their prerogatives and have transferred only limited competencies to the EU. Seen as an institution, the ECB (as opposd to the Eurosystem) conducts only a small amount of market operations, has no independent decision-making authority, and has no single finance ministry with which to coordinate policy. The national central banks, which together with the ECB make up the ESCB[16], have considerable powers and are involved in financial markets to a much larger degree than the ECB itself. The situation of the Fed is not comparable although the Fed also operates in a decentralised fashion. While the Federal Reserve Board in Washington also does not conduct market operations, it has outsourced all of this work to a single Reserve Bank, that of New York. In some of the more recent policy programmes, the Fed was able to obtain fiscal backstops for the credit risk it took in asset purchase programmes.

The financial crisis has highlighted the shortcomings of this institutional arrangement. National interests diverge between countries with large external lending positions, those with large external deficits, and those who had a large financial intermediation sector. It is often said that the monetary union of the euro area is incomplete because it is not flanked by a fiscal union. While true, this statement is not in fact particularly meaningful. Incompleteness must be measured against a benchmark, and a fiscal union is an obvious but not necessary one. So far at least, the monetary union has not turned out to be dysfunctional which arguably is a more important concern than incompleteness.

In this setting, the ECB is faced with the task of acting in a comprehensive manner while having been carefully designed not to be able to act other than in very narrow ways. This classical dilemma has led to an uncomfortable outcome which shows in the terms ‘input legitimacy’ and ‘output legitimacy’ used in public communications [72][17]. The concept of ‘output legitimacy’ is periously close to ‘the end justifies the means’. Karl Marx already remarked that if the end justifies the means, then the end is unjustified[18]. In the specific context of examining the actions of a central bank, ‘output legitimacy’ is

[16] European System of Central Banks.
[17] The terms are taken from speeches by former board member Jörg Asmussen but the concepts appear in a number of ECB communications including the submissions to the German Constitutional Court.
[18] ‘Wenn der Zweck die Mittel heiligt, dann ist der Zweck unheilig.’

also logically meaningless because only one possible output, namely the present state, can be observed. Any alternative outcomes are necessarily pure conjecture because only one currency area with the specific features of the Eurozone exists. Legitimacy, however, cannot result from conjecture. For this reason, 'output legitimacy' is not a valid framework for judging the admissibility of specific policy actions.

The issue came to the fore in 2020 when the German constitutional court ruled the ECB's Public Sector Purchase Programme to be unlawful (ultra vires) on the grounds that it had not been properly justified ex-ante: 'it would have been incumbent upon the ECB to weigh these effects and balance them, based on proportionality considerations, against the expected positive contributions to achieving the monetary policy objective the ECB itself has set… For this lack of balancing and lack of stating the reasons informing such balancing, the ECB decisions at issue violate Art. 5(1) second sentence and Art. 5(4) TEU and, in consequence, exceed the monetary policy mandate of the ECB deriving from Art. 127(1) first sentence TFEU.' (nos. 176f in [17]). The ECB is not subject to the jurisdiction of the German constitutional court and EU law is assumed to be the exclusive jurisdiction of the European Court of Justice. That court did not share this opinion when it ruled, at the request of the German court, on PSPP in 2018 (ECJ case C-493/17). While questioning the lack of an ex-ante justification of PSPP, the German court did not see a violation of EU Treaties in the execution of PSPP ex-post.

At the time of writing, this particular conflict appears to have been resolved by forwarding additional information about the decision-making process regarding PSPP to the German government and parliament. As discussed above, the issue of communicating not only monetary policy decisions, but also their rationale, is evolving in the global central banking community.